

Table 4. Low-entropy code-domain stress test on MBPP using a pretrained CodeLlama checkpoint. Results are reported on 50 MBPP tasks with 10 samples per task. Detection statistic denotes the mean detector output (z-score when available).

Method	pass@1	pass@10	Parse Rate	Detection Rate	Mean Detection Statistic	Interpretation
Vanilla	0.076	0.420	0.328	0.000	0.00	Baseline low-entropy reference point.
KGW	0.062	0.340	0.256	0.804	$z = 7.42$	Clear utility drop with strong keyless detectability.
Unigram	0.036	0.320	0.298	0.816	$z = 8.30$	Largest utility drop among measured methods, with strong detectability.
Hybrid	0.076	0.420	0.290	0.454	$z = 2.29$	Best utility preservation among detectable methods; moderate detectability.
CGW	0.068	0.420	0.308	0.000	$z = -0.75$	Near-baseline utility with negligible keyless detectability in this setting.

Note. The unwatermarked baseline has mean next-token entropy 0.93 bits and mean top-1 probability 0.84, confirming that MBPP/CodeLlama is a substantially lower-entropy regime than the main LFQA setting. In this regime, utility degradation is better reflected by diversity-sensitive metrics such as pass@k than by text similarity alone.